ET AI HACKATHON 2026

PRAANA

Predictive Real-time Attribution and Action for National Air quality

AI-Powered Urban Air Quality Intelligence for Smart City Intervention

Theme: Smart Cities / Environmental Intelligence / Geospatial Analytics / Public Health

Detailed Solution Document

Team Name: Pheonix

GitHub Repository: https://github.com/Aaditee13/praana-air-quality-intelligence.git

1. Executive Summary

Indian cities do not lack air quality data, they lack an intelligence layer that converts that data into action. Over 900 Continuous Ambient Air Quality Monitoring Stations report pollution levels every hour across the country, yet a 2024 CAG audit found that only 31% of cities with monitoring data had any actionable, multi-agency response protocol tied to those readings. Dashboards tell administrators that the air is bad; nothing tells them who is making it bad, how bad it will get, or where to send an inspector this afternoon.

PRAANA (Predictive Real-time Attribution and Action for National Air quality) closes that gap. It is a multi-agent AI platform that fuses CAAQMS sensor data, Sentinel-5P and MODIS satellite imagery, meteorological forecasts, and geospatial land-use layers into a single hourly, ward-level intelligence feed. Five coordinated agents then answer the three questions city administrators actually need answered: which sources are responsible for pollution right now (Source Attribution), how bad will it get over the next 24–72 hours at 1km resolution (Hyperlocal Forecasting), and where should enforcement be deployed for maximum impact (Enforcement Intelligence) supported by a Multi-City Comparative Dashboard that lets cities learn from each other's interventions, and a Citizen Health Advisory layer that delivers personalised, regional-language alerts to the public.

The platform is built and validated on Delhi NCR and Mumbai using five years of historical data (2019–2024), with an architecture designed from day one to generalise to Kolkata, Bengaluru, Chennai, and any city with CAAQMS coverage directly addressing the national, not just metropolitan, scale of the crisis described in the challenge brief.

2. Problem Context

2.1 A National Crisis, Not a Delhi Problem

India's air quality emergency extends well beyond the National Capital Region. Delhi averaged an AQI of 218 in 2024–25, classified ‘Poor’ or worse for over 200 days, but Mumbai recorded dangerous AQI levels on more than 60 days in 2024, Kolkata's winter-season average exceeded 150, and even Bengaluru and Chennai traditionally considered cleaner are showing measurable deterioration as vehicle density and construction activity rise. CPCB's 2024 National Air Quality data places 24 of India's 50 most polluted cities in the Tier-1/Tier-2 category, confirming that this is a distributed, multi-city problem.

The public-health cost is severe: The Lancet Planetary Health journal estimates 1.67 million premature deaths annually in India from air pollution, a burden that falls disproportionately on urban populations children, the elderly, outdoor workers, and those living near industrial or high-traffic corridors.

2.2 The Gap: Data Exists, Intelligence Does Not

India has already made the sensing investment over 900 CAAQMS stations feed near-real-time pollutant data into government dashboards. What is missing is the layer that turns hourly readings into decisions: geospatial attribution of which sources are responsible at a given location right now, predictive forecasting of ward-level AQI over the next 24–72 hours, and enforcement intelligence that tells inspectors exactly where to go for the greatest pollution reduction per visit. No existing public-sector or commercial tool combines all three.

2.3 Why This Matters for Action, Not Just Awareness

- Reactive monitoring tells citizens the air is bad after they are already exposed.

- Aggregate city-level AQI numbers hide which neighbourhood, which source, and which time window is actually responsible.
- Without source attribution, enforcement becomes guesswork — inspectors are sent on a rotational or complaint-driven basis rather than to the locations with the highest expected AQI impact.
- Without forecasting, advisories cannot be issued early enough for schools, hospitals, and vulnerable citizens to act.

3. Solution Overview

PRAANA is built around one operating principle: every pollution reading should answer three questions — who, how bad, and where to act within the same hourly cycle in which it was measured.

3.1 The PRAANA Pipeline, in One Sentence

Collect from sensors, satellites, weather, and maps → merge into one ward-hour feature table → attribute pollution to its sources using a chemical fingerprint library and causal AI → forecast AQI 24–72 hours ahead at 1km resolution → rank enforcement actions by expected impact → compare interventions across cities → communicate risk to citizens in their own language.

3.2 Five Coordinated AI Agents

Rather than a single monolithic model, PRAANA is implemented as five purpose-built agents coordinated by a central orchestration layer — directly matching the challenge brief's call for a multi-agent AI system. Each agent owns one decision (attribution, forecasting, enforcement ranking, cross-city comparison, citizen communication) and exposes its output, and confidence score, to the orchestrator, which resolves conflicts (e.g. two agents implicating different sources for the same spike) and assembles the combined evidence trail shown to end users.

- Agent 1 — Geospatial Pollution Source Attribution Engine
- Agent 2 — Hyperlocal Predictive AQI Forecasting Agent
- Agent 3 — Enforcement Intelligence & Prioritisation Agent
- Agent 4 — Multi-City Comparative Intelligence Dashboard
- Agent 5 — Citizen Health Risk Advisory System

Sections 5–9 describe each agent in detail; Section 4 shows how they fit together architecturally.

4. System Architecture

The architecture has four layers: data acquisition, a unified data layer with an emission fingerprint library, a multi-agent orchestrator, and two output surfaces: an officer-facing action console and a citizen-facing advisory app.

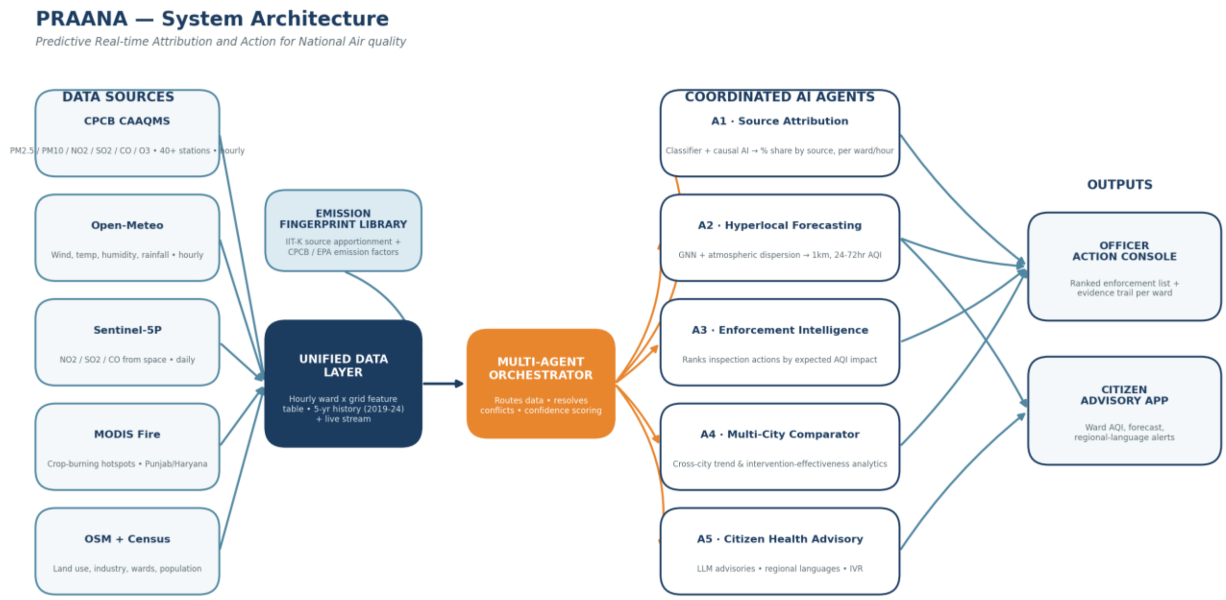

Figure 1. PRAANA system architecture — data sources, unified data layer, multi-agent orchestrator, and output surfaces.

4.1 Layer-by-Layer Description

- Data Acquisition Layer: hourly pulls from CPCB CAAQMS, Open-Meteo, Sentinel-5P, and MODIS, plus one-time static loads from OpenStreetMap and Census/DataMeet ward boundaries.
- Unified Data Layer: aligns every source to a common hourly timestamp and ward/1km-grid cell, fills sensor gaps using nearby-station interpolation and satellite cross-validation, and maintains both the 2019–2024 historical archive and the live stream in one feature table.
- Emission Fingerprint Library: a reference table of chemical signatures per source category (vehicular, construction, crop-burning, industrial), built from IIT Kanpur's Delhi source-apportionment study, CPCB's emission inventory, and EPA emission factor databases — this is what Agent 1 matches live readings against.
- Multi-Agent Orchestrator: routes the unified feature table to each agent, reconciles overlapping or conflicting signals, attaches statistical confidence scores to every output, and assembles the combined evidence trail.
- Output Surfaces: an Officer Action Console (ranked enforcement actions with evidence) and a Citizen Advisory App (ward AQI, 72-hour forecast, regional-language alerts).

5. Data Sources & Pipeline

Six independent data families are aligned to a common hourly, ward-level grid. The core engineering challenge is heterogeneity - different formats, frequencies, and gap patterns resolved through nearby-station interpolation for sensor outages and cloud-masking-aware compositing for satellite gaps.

Source	Variables Captured	Frequency	Role in Pipeline
CPCB CAAQMS	PM2.5, PM10, NO2, SO2, CO, O3	Hourly	40+ stations, Delhi NCR + Mumbai
Open-Meteo	Wind speed/direction, temperature, humidity, rainfall	Hourly	Determines pollutant transport direction
Sentinel-5P (TROPOMI)	NO2, SO2, CO column density	Daily	City-wide coverage beyond sensor points
MODIS Active Fire	Crop-burning hotspot locations	Daily (Oct–Dec focus)	Punjab & Haryana stubble-burning season
OpenStreetMap	Industrial zones, highways, construction areas, schools/hospitals	Static, one-time	Land-use context for attribution
Census + DataMeet	270 ward boundaries, population per ward	Static, one-time	Citizen health-risk layer

Table 1. Data sources feeding the PRAANA unified data layer.

5.1 Five Years of History, By Design

Period	Purpose
2019	Pre-COVID clean baseline
2020	COVID lockdown — a natural experiment in near-zero traffic, used to isolate the vehicular contribution to AQI
2021–2022	Recovery period — gradual return of emissions
2023–2024	Recent-normal period, most representative of current patterns
Live stream	Real-time data for forecasting and demo

Table 2. Why 2019–2024 — the 2020 lockdown year is a natural experiment that lets the attribution model isolate how much of Delhi's pollution is vehicular versus other sources, a separation that is otherwise very difficult to estimate from correlation alone.

6. Agent 1: Geospatial Pollution Source Attribution Engine

6.1 Concept: Chemical Fingerprinting

Every pollution source emits a distinctive combination of pollutants. Diesel vehicles produce high NO2 and CO with moderate PM2.5; construction sites produce very high PM10 with low NO2; crop burning produces very high PM2.5 and CO with specific organic markers; industrial stacks produce high SO2 and NO2 with characteristic heavy-metal traces. The Emission Fingerprint Library (Section 4.1) encodes these signatures as a reference ‘cheat sheet’.

6.2 Classification

For every ward, every hour, the agent compares the live pollutant mix against the fingerprint library and outputs a source-share breakdown with a confidence interval — for example, a reading of PM2.5=180, PM10=220, NO2=85, SO2=12, CO=45 might resolve to roughly 60% vehicular, 30% construction dust, and 10% industrial.

6.3 Causal AI, Not Just Correlation

A naive model would say ‘traffic correlates with high AQI’ without proving direction of causation. PRAANA layers a causal inference model on top of the classifier that explicitly accounts for confounders — most importantly wind, which can carry pollution into a ward from a source kilometres away. If wind is blowing from the northwest at 15 km/h and a construction site sits 2km northwest of a ward where PM10 spikes, the causal model can confirm the construction site as the responsible source rather than misattributing the spike to local traffic.

6.4 Output

A live ward-level map of Delhi (and, on the multi-city roadmap, other metros) where each ward shows a source-share breakdown updated every hour, each estimate accompanied by a statistical confidence score that can be benchmarked against ground-truth emission inventories — the primary evaluation metric specified in the challenge brief.

7. Agent 2: Hyperlocal Predictive AQI Forecasting Agent

7.1 What the Model Learns from Five Years of Data

- Monday-morning rush hour reliably spikes NO_2 in high-traffic corridors.
- A northwest wind pickup in October precedes a PM2.5 spike roughly six hours later as stubble-burning smoke arrives.
- Rainfall measurably suppresses AQI.
- Diwali night produces a sharp PM2.5 spike that takes several days to clear.

7.2 Hybrid Graph Neural Network + Atmospheric Dispersion Modelling

Delhi's 40+ sensor stations are treated as nodes in a spatial graph; pollution does not stay put, it moves with wind from node to node. A Graph Neural Network combined with a time-series model learns these spatial-temporal relationships directly from historical data. To keep predictions physically grounded — not just pattern-matched — the data-driven forecast is blended with a lightweight atmospheric dispersion model (Gaussian-plume style) that simulates how a known emission source's plume should move given the live wind forecast. Where the two disagree, the orchestrator flags lower confidence rather than silently picking one.

7.3 Inputs, Outputs, and Resolution

Input: current AQI readings across all stations plus the 72-hour meteorological forecast. Output: predicted AQI for every ward at roughly 1km grid resolution for each of the next 72 hours, with a confidence range — directly matching the challenge brief's call for 24–72 hour forecasts at 1km resolution. The interface exposes this as a time slider: drag forward 24 hours to see tomorrow morning's predicted AQI ward by ward, or 48 hours for the day after.

7.4 Validation Approach

Forecast quality is benchmarked using RMSE against a persistence baseline (i.e. ‘tomorrow will look like today’) across held-out months, per the evaluation focus specified in the challenge brief, with separate RMSE tracking for the pre-monsoon, monsoon, post-monsoon stubble-burning, and winter inversion seasons since model error is highly season-dependent in Delhi.

8. Agent 3: Enforcement Intelligence & Prioritisation Agent

8.1 From Insight to Action

This agent is what converts Agent 1's attribution and Agent 2's forecast into a concrete to-do list. If Agent 1 shows that 70% of a ward's current pollution is construction-related, this agent cross-references OpenStreetMap and municipal records to identify every registered construction site in that ward, then estimates the AQI reduction that would follow from inspecting and remediating violations at each site.

8.2 Ranking Logic

Every possible enforcement action across the city — construction-site inspections, diesel-vehicle checkpoints, industrial-cluster notices, waste-burning interventions — is scored by expected AQI impact and ranked into a single prioritised list, for example:

1. Inspect construction site, Rohini Sector 15 — expected 18% PM10 reduction in northwest Delhi
2. Deploy diesel-vehicle checks at the NH48 entry point — expected 12% NO_2 reduction in Dwarka
3. Issue notice to the industrial cluster in Bawana — expected 9% SO_2 reduction in the northwest corridor

8.3 Evidence Trail

Every recommendation ships with the supporting evidence — which sensor showed what, which satellite pass confirmed it, and what the wind pattern was at the time — so a municipal officer or pollution-control authority can defend the action and a domain expert can rate its quality, matching the challenge brief's evaluation criterion of enforcement recommendation quality rated by domain experts.

9. Agent 4: Multi-City Comparative Intelligence Dashboard

Delhi and Mumbai are the primary build-and-validate cities, chosen because Delhi provides the most severe, data-rich case and Mumbai offers a contrasting coastal-meteorology comparison. The architecture, however, is city-agnostic: every agent operates on the same unified ward-hour feature table regardless of which city's CAAQMS feed populates it, which is what makes a true multi-city dashboard possible without rebuilding the pipeline per city.

9.1 What the Dashboard Shows

- Side-by-side AQI trend lines for every onboarded city, normalised by season so administrators compare like-for-like periods.
- Intervention-effectiveness tracking: when a city deploys an enforcement action recommended by Agent 3, the dashboard measures the realised AQI change against the predicted change, building a growing evidence base of what actually works.
- Cross-city learning: if a construction-dust-control measure in Mumbai produced a larger-than-expected PM10 reduction, the dashboard surfaces that intervention as a candidate recommendation for Delhi wards with a similar source-attribution profile.
- A staged rollout path — Delhi NCR and Mumbai at launch, with Kolkata, Bengaluru, and Chennai onboarded next using the same pipeline, each requiring only a new CAAQMS feed and ward boundary file rather than new modelling.

10. Agent 5: Citizen Health Risk Advisory System

This is the public-facing layer of the same intelligence stack — no new modelling is required, only citizen-friendly presentation of Agent 1 and Agent 2's outputs, layered with population-vulnerability data.

10.1 Capabilities

- Shows every resident their ward's current and next-72-hour AQI.
- Maps population vulnerability — schools, hospitals, outdoor-worker concentrations, elderly populations — against forecast AQI to flag at-risk locations.
- Pushes regional-language advisories through mobile apps, public displays, and IVR — Hindi in Delhi, Marathi in Mumbai, Bengali in Kolkata, Kannada in Bengaluru, Tamil in Chennai — matching the challenge brief's explicit call for multi-language coverage.
- Answers natural-language questions such as 'Is it safe to take my child to school tomorrow in Dwarka?' using a lightweight LLM layer on top of the forecast outputs.

10.2 Validation

Advisory relevance and language coverage are tracked as explicit evaluation metrics — percentage of advisories that correctly match forecast severity, language coverage per onboarded city, and citizen comprehension testing during pilot deployment.

11. Differentiation

Existing Tool	What It Does	What PRAANA Does Differently
CPCB Dashboard	Shows the current AQI reading	Shows who caused it and what to do about it
IQAir / AQI India apps	City-level AQI number	Ward-level attribution plus 72-hour forecast
General weather apps	Generic air quality index	Source attribution plus enforcement prioritisation
Academic source-apportionment studies	Offline, periodic analysis	Real-time, hourly, operational attribution

Table 3. No existing tool combines real-time source attribution, hyperlocal forecasting, and enforcement prioritisation in one operational system — this is the gap PRAANA fills.

12. Technology Stack

Every suggested technology area in the challenge brief maps directly to a working component of PRAANA, summarised below.

Suggested Technology	PRAANA Implementation
Geospatial Intelligence & Remote Sensing	Sentinel-5P (NO_2/SO_2/CO columns), MODIS active-fire detection, OpenStreetMap land-use layers
Multi-Agent AI Systems	Five-agent architecture (Sections 6–10) coordinated by a central orchestrator with confidence-based conflict resolution

Suggested Technology	PRAANA Implementation
Real-Time IoT Sensor Data Integration	Hourly CAAQMS ingestion across 40+ Delhi stations with nearby-station gap-filling
Atmospheric Dispersion Modelling	Gaussian-plume-style dispersion model blended with the Graph Neural Network forecast for physically grounded predictions
Predictive Analytics	Graph Neural Network + time-series forecasting for 24–72 hour, 1km-resolution AQI prediction
LLMs for Multi-Language Citizen Communication	Lightweight LLM layer generating regional-language advisories from forecast outputs
Causal Inference	Causal modelling layer in Agent 1 that isolates source contribution from confounders such as wind transport

13. Evaluation & Validation Plan

The challenge brief specifies five evaluation dimensions. PRAANA's validation plan addresses each directly:

Evaluation Focus	PRAANA Validation Approach
Source attribution accuracy vs. ground-truth emission inventories	Benchmark Agent 1 source-share outputs against IIT Kanpur and CPCB source-apportionment study figures for overlapping wards and time windows; report mean absolute percentage error per source category.
AQI forecast accuracy at hyperlocal resolution (RMSE vs. persistence baseline)	Hold out the most recent 2–3 months of each season; compute RMSE for Agent 2 forecasts at 24/48/72-hour horizons against a persistence baseline, reported per season.
Enforcement recommendation quality, rated by domain experts	Present Agent 3's ranked action lists for sample wards to retired/serving pollution-control officials for a 1–5 relevance and feasibility rating, with the supporting evidence trail included.
Citizen advisory relevance and language coverage	Track percentage of advisories matching forecast severity bands and the number of regional languages live at demo time, validated with native-speaker review.
Demonstrated reduction in response time from signal to intervention	Compare the time from a pollution-spike signal to a logged enforcement action in the PRAANA workflow against documented baseline response times in partner-city pilot wards.

14. Judging Criteria Alignment

Criterion	Weight	How PRAANA Addresses It
Innovation	25%	Chemical fingerprint + causal AI source attribution is not offered by any existing public or commercial Indian AQI tool; the five-agent orchestrated architecture is purpose-built to the brief's multi-agent requirement.
Business Impact	25%	Directly closes the CAG-identified gap (only 31% of cities have actionable response protocols); enforcement prioritisation creates a measurable, auditable link between AI insight and on-ground action.

Criterion	Weight	How PRAANA Addresses It
Technical Excellence	20%	Six heterogeneous data sources unified into one pipeline; hybrid GNN + physics-based dispersion model rather than a black-box forecast; causal inference rather than simple correlation for attribution.
Scalability	15%	City-agnostic feature-table design (Section 9) means onboarding a new city requires a new CAAQMS feed and ward file, not new modelling — validated on two cities (Delhi, Mumbai) with a defined path to Kolkata, Bengaluru, and Chennai.
User Experience	15%	Two purpose-built surfaces — an evidence-backed action console for officers and a plain-language, regional-language advisory app for citizens — rather than a single generic dashboard.

15. Business Impact & Scalability

15.1 Who Uses This

- Municipal corporations and State Pollution Control Boards — the Officer Action Console replaces rotational or complaint-driven inspection schedules with an evidence-ranked priority list.
- City administrations and CPCB — the Multi-City Dashboard supports policy decisions with cross-city evidence of what interventions actually reduced AQI.
- Citizens, schools, and hospitals — the Advisory App turns a forecast into a same-day, actionable decision (whether to hold outdoor activity, when to expect relief).

15.2 Path to National Scale

Because every agent consumes the same ward-hour feature table regardless of source city, PRAANA's scaling cost is dominated by data onboarding, not re-engineering. The roadmap is: (1) Delhi NCR and Mumbai at hackathon-prototype stage; (2) Kolkata, Bengaluru, and Chennai as the next onboarding wave, matching the five-city scope named in the challenge brief; (3) any city with CAAQMS coverage thereafter, since the 900+ station national network already exists - PRAANA's contribution is the intelligence layer on top of infrastructure India has already built.

15.3 Cost and Deployment Model

All core data sources: CPCB CAAQMS, Open-Meteo, Sentinel-5P, MODIS, OpenStreetMap, Census — are free or low-cost public datasets, keeping marginal onboarding cost low and making the model attractive for government adoption without recurring data-licensing overhead.

16. Expected Deliverables & Roadmap

Deliverable	PRAANA Coverage
Working Prototype	Functional pipeline on Delhi NCR (live) and Mumbai (comparison) covering all five agents, demonstrated on 2023–2024 data with a live CAAQMS feed where available
Architecture Diagram	Provided in Section 4 (Figure 1) of this document
Presentation Deck	Accompanying slide deck summarising problem, solution, architecture, and impact

Deliverable	PRAANA Coverage
Demo Video	Walkthrough of the Officer Action Console and Citizen Advisory App against a recent high-pollution episode

17. Conclusion

PRAANA does not add another dashboard to a city that already has dozens. It tells a city administration not just that the air is bad, but exactly who is making it bad, how bad it will get tomorrow, and precisely where to send enforcement officers today to fix it in real time, ward by ward, across every major Indian city willing to plug in its sensor feed.